

Fundamentals of Data Visualization

K Arnold, based on [IntroDS.org](https://introds.org)

Review

What do each of the following operations do in RStudio?

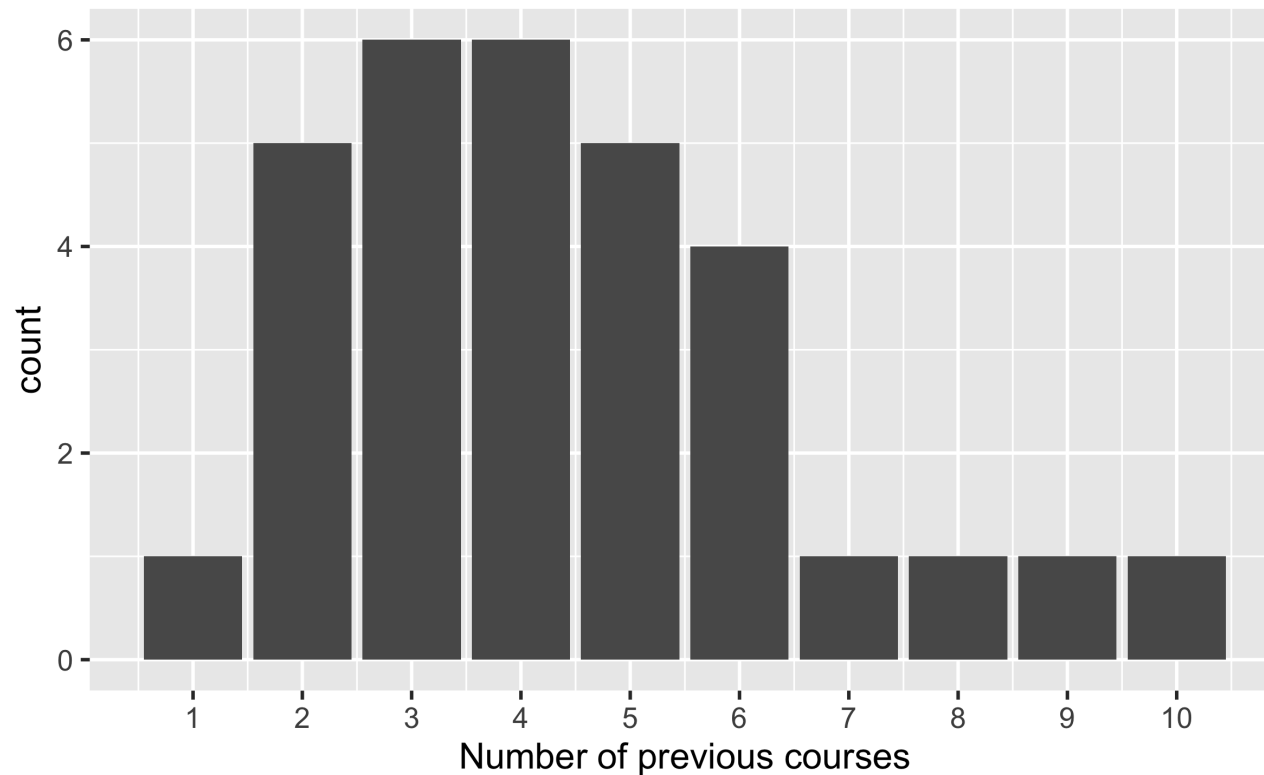
- Save
- Run Cell
- Knit
- Commit
- Push

Think-Pair-Share.

From the survey (Quiz 1)

covid	n
I'd be in favor of using such examples.	10
I'm neutral.	17
I'd prefer something different but I'd be ok with it.	2
I feel strongly that we should not use such examples.	2

```
quiz_responses %>%  
  mutate(num_courses = str_count(courses, "[0-9][0-9][0-9]")) %>%  
  ggplot(aes(x = num_courses)) +  
    geom_bar() +  
    scale_x_continuous(breaks = 1:10) +  
    labs(x = "Number of previous courses")
```



Q & A

| Can we edit directly on GitHub?

Technically yes, but GitHub won't show plots, documentation, etc., so don't.

| What programming languages?

- We're using R (**#rstats**) because it has a big community, it's popular in industry, and is good pedagogically.
- Other players: Python (good at many things), Stata / SPSS / JMP (popular in some disciplines), Tableau / Microsoft PowerBI (popular in business)

| Are post-class quizzes part of grade?

They count towards "Prep and Participation" (10%).

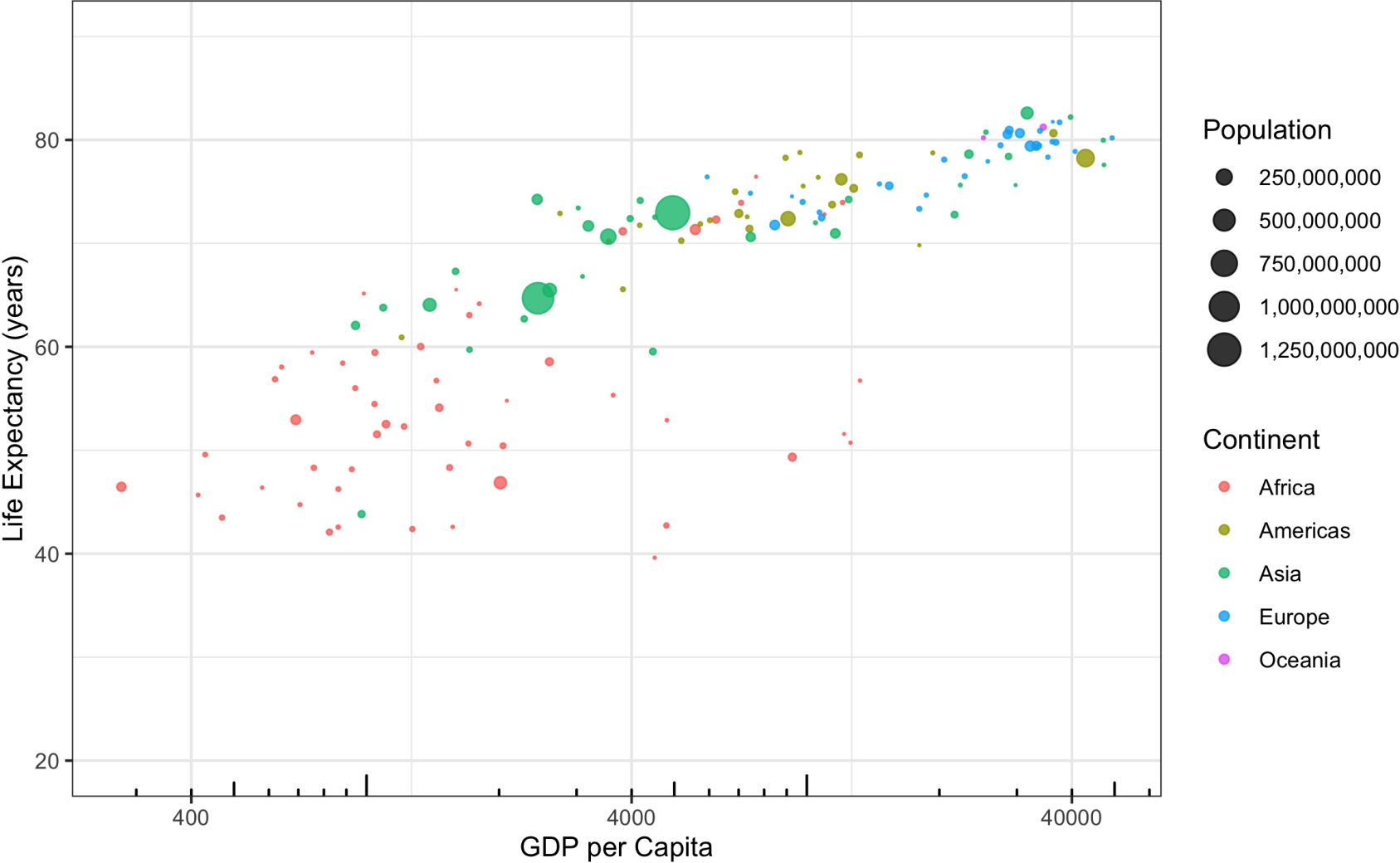
 **knit**

 **commit**

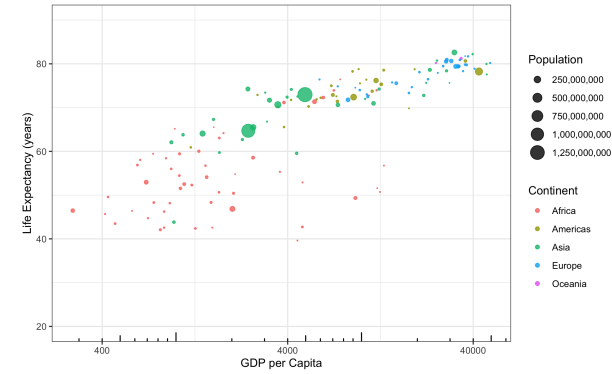
 **push**



We'll make this chart

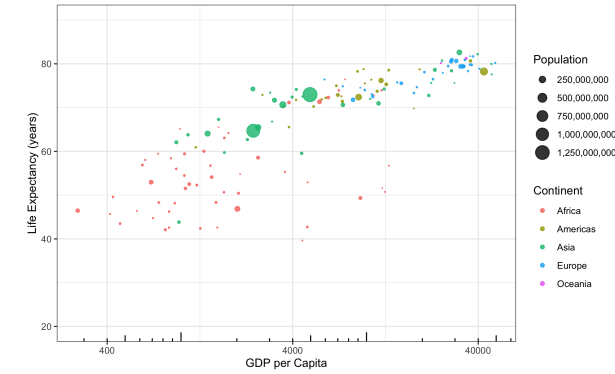


Composing a plot



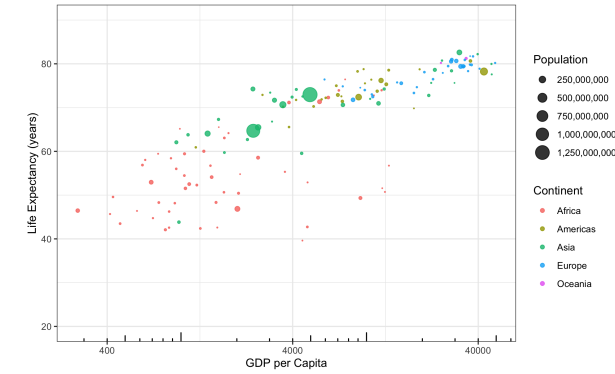
Composing a plot

- What's the data? "Each row is a _"



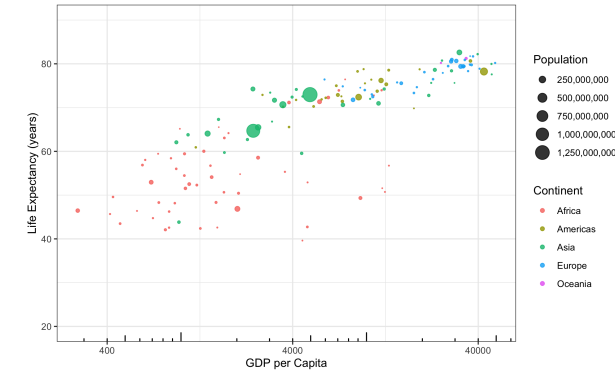
Composing a plot

- What's the data? "Each row is a _"
- What is the coordinate system? (What's **x** and **y**?)



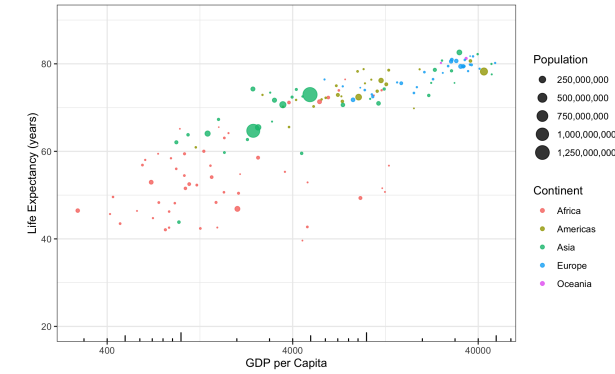
Composing a plot

- What's the data? "Each row is a _"
- What is the coordinate system? (What's **x** and **y**?)
- What graphical symbols are used? (dot? bar? line?)



Composing a plot

- What's the data? "Each row is a _"
- What is the coordinate system? (What's **x** and **y**?)
- What graphical symbols are used? (dot? bar? line?)
- What data variables are mapped to what visual cues (aesthetics)?
 - What *scales* are used? (Any transformations?)
 - What *guides* are shown? (What labels for values?)
- What labels and annotations are added?



gapminder

```
## # A tibble: 1,704 × 6
##   country      continent  year lifeExp      pop gdpPercap
##   <fct>        <fct>    <int>   <dbl>    <int>    <dbl>
## 1 Afghanistan Asia      1952    28.8  8425333    779.
## 2 Afghanistan Asia      1957    30.3  9240934    821.
## 3 Afghanistan Asia      1962    32.0 10267083    853.
## 4 Afghanistan Asia      1967    34.0 11537966    836.
## 5 Afghanistan Asia      1972    36.1 13079460    740.
## 6 Afghanistan Asia      1977    38.4 14880372    786.
## # ... with 1,698 more rows
```

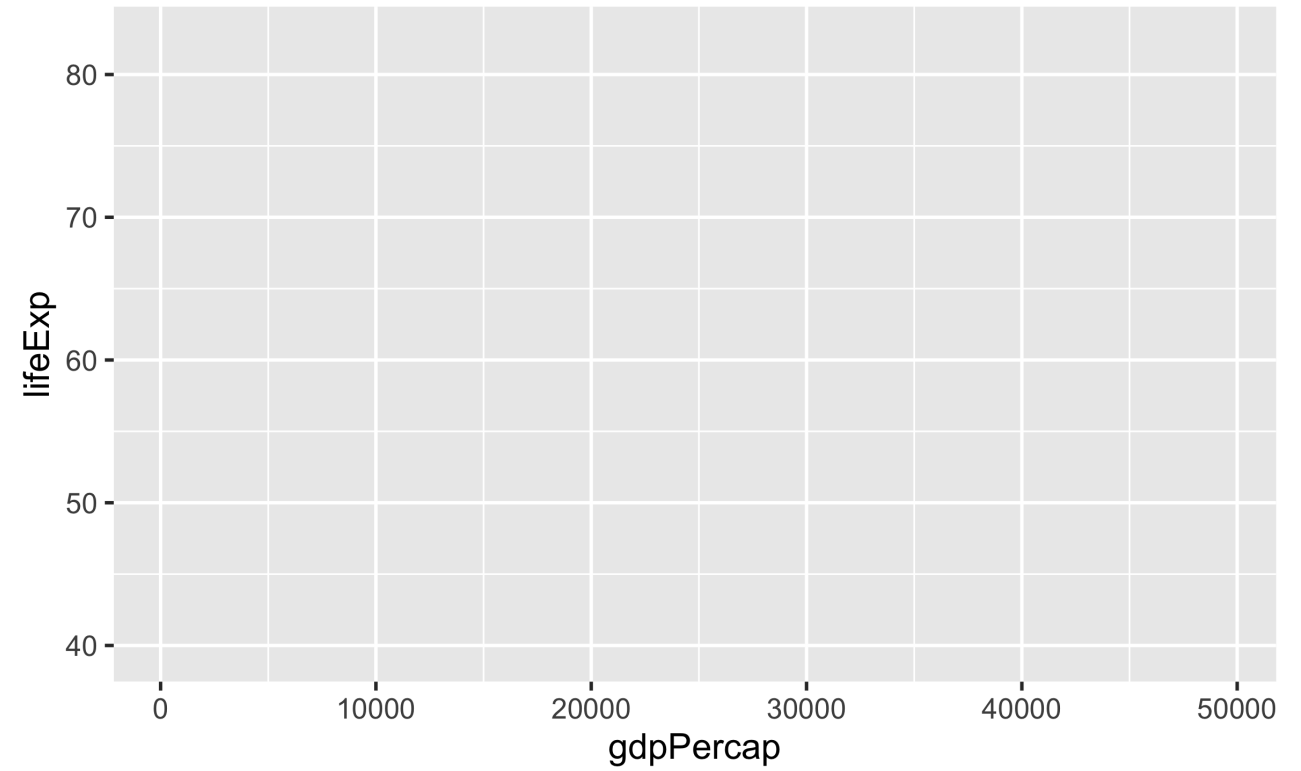
```
gapminder %>%
```

```
  filter(year == 2007)
```

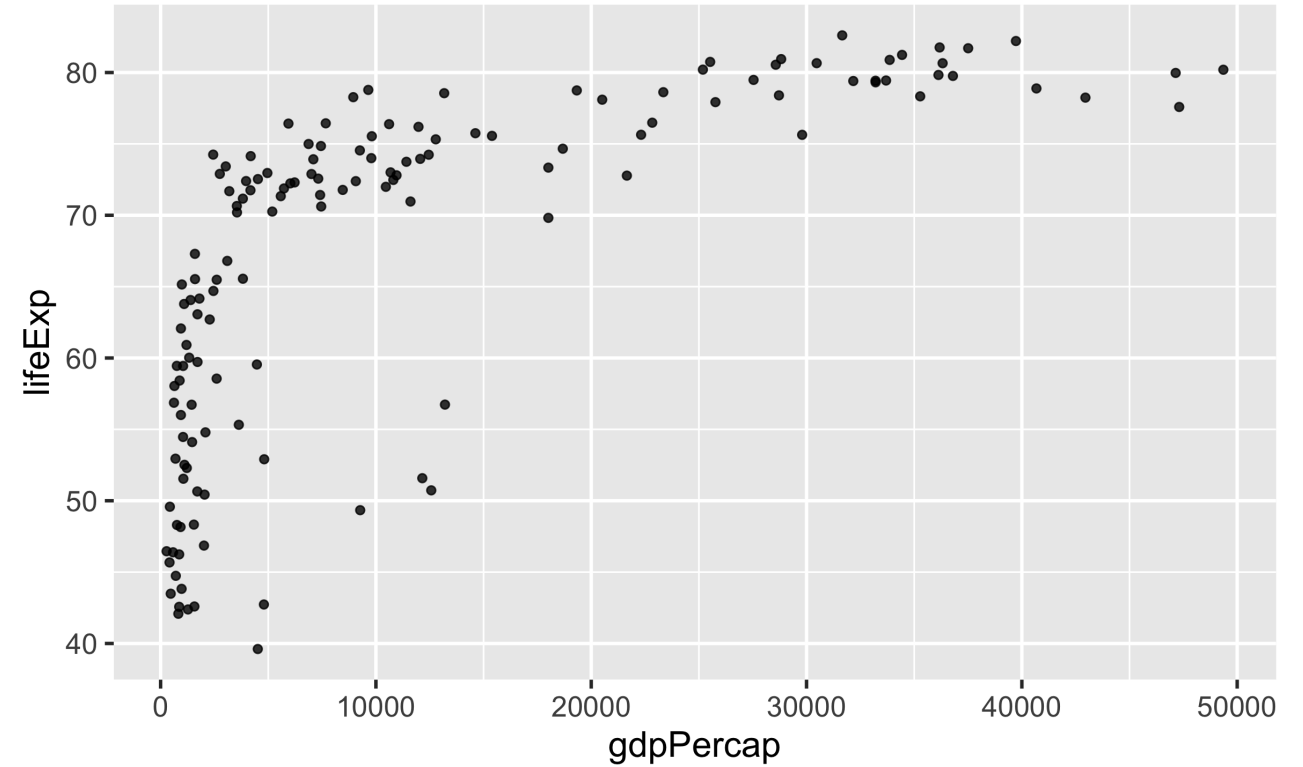
```
## # A tibble: 142 × 6
##   country      continent year lifeExp      pop gdpPercap
##   <fct>         <fct>   <int>   <dbl>    <int>    <dbl>
## 1 Afghanistan Asia       2007    43.8 31889923    975.
## 2 Albania      Europe    2007    76.4 3600523   5937.
## 3 Algeria      Africa    2007    72.3 33333216   6223.
## 4 Angola       Africa    2007    42.7 12420476   4797.
## 5 Argentina    Americas  2007    75.3 40301927  12779.
## 6 Australia    Oceania   2007    81.2 20434176  34435.
## # ... with 136 more rows
```

```
gapminder %>%  
  filter(year == 2007) %>%  
  ggplot()
```

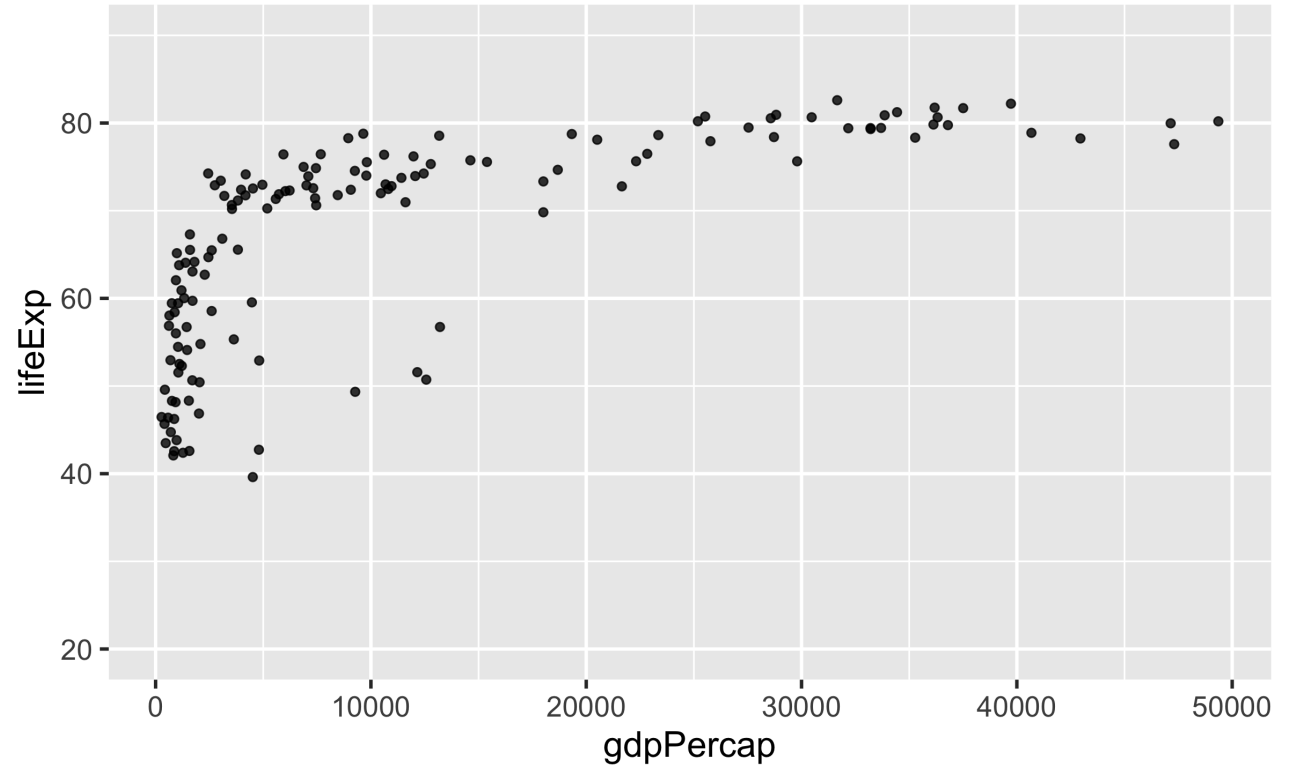
```
gapminder %>%  
  filter(year == 2007) %>%  
  ggplot() +  
    aes(x = gdpPercap, y = lifeExp)
```



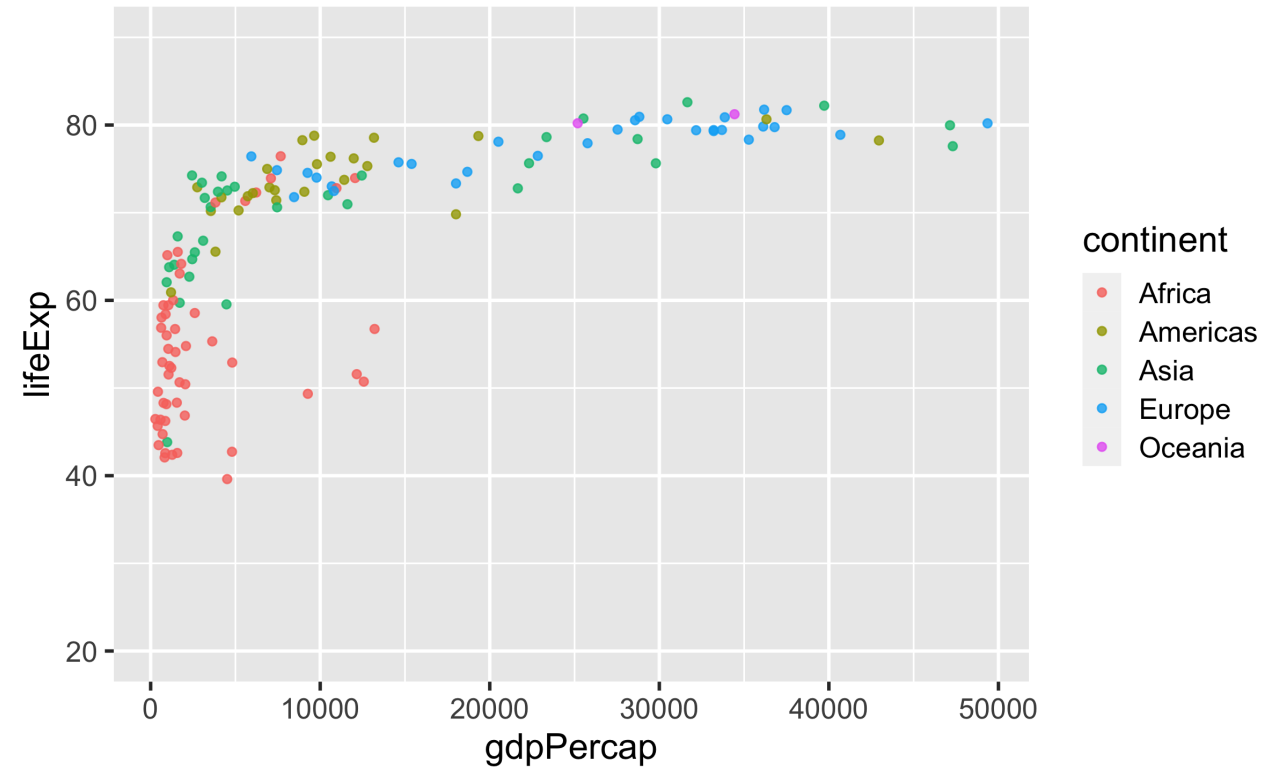
```
gapminder %>%  
  filter(year == 2007) %>%  
  ggplot() +  
    aes(x = gdpPercap, y = lifeExp) +  
    geom_point(alpha = .8)
```



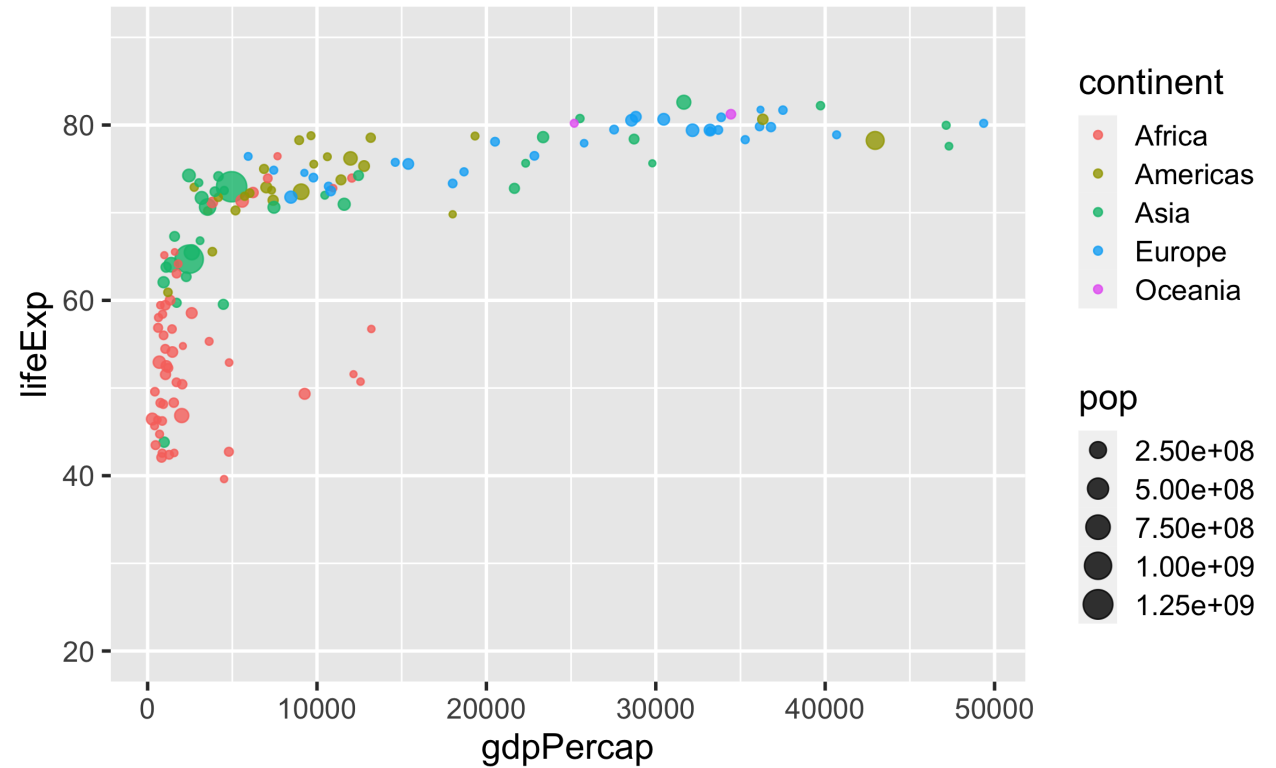
```
gapminder %>%  
  filter(year == 2007) %>%  
  ggplot() +  
    aes(x = gdpPercap, y = lifeExp) +  
    geom_point(alpha = .8) +  
    coord_cartesian(ylim = c(20, 90))
```



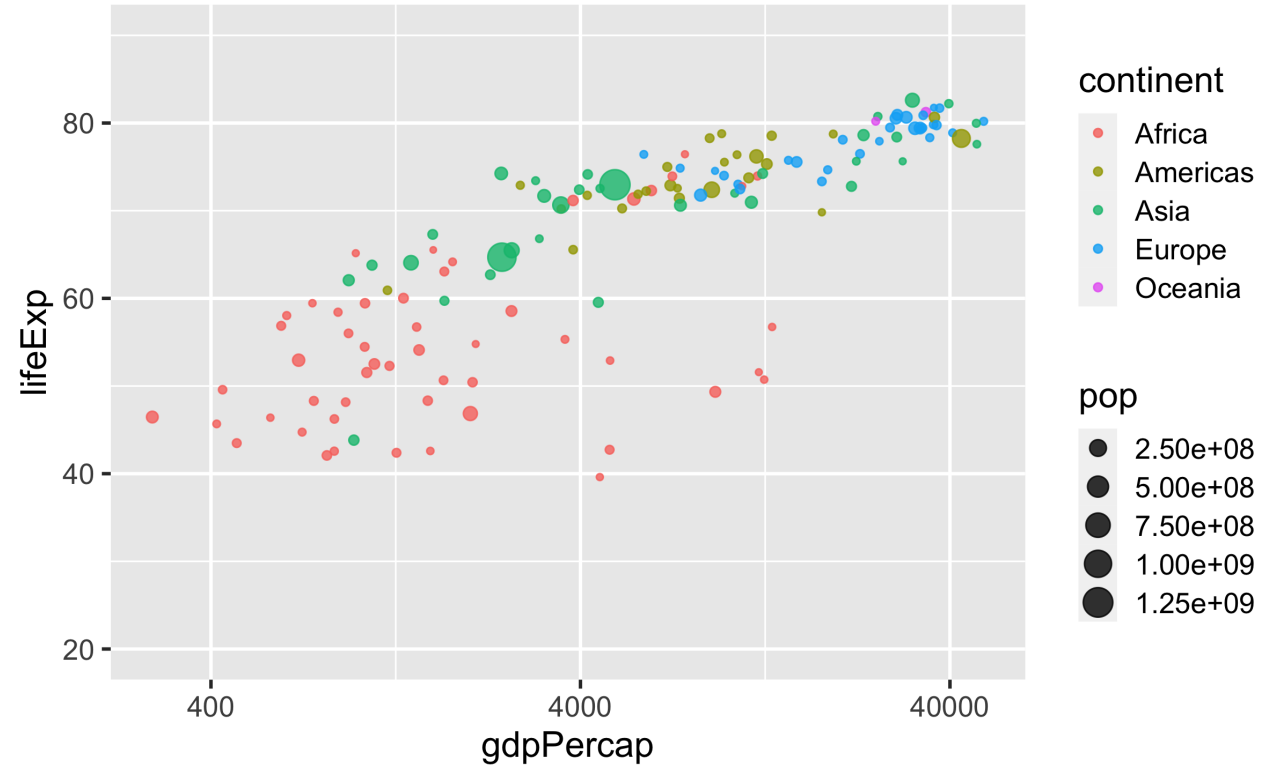
```
gapminder %>%  
  filter(year == 2007) %>%  
  ggplot() +  
    aes(x = gdpPercap, y = lifeExp) +  
    geom_point(alpha = .8) +  
    coord_cartesian(ylim = c(20, 90))  
    aes(color = continent)
```



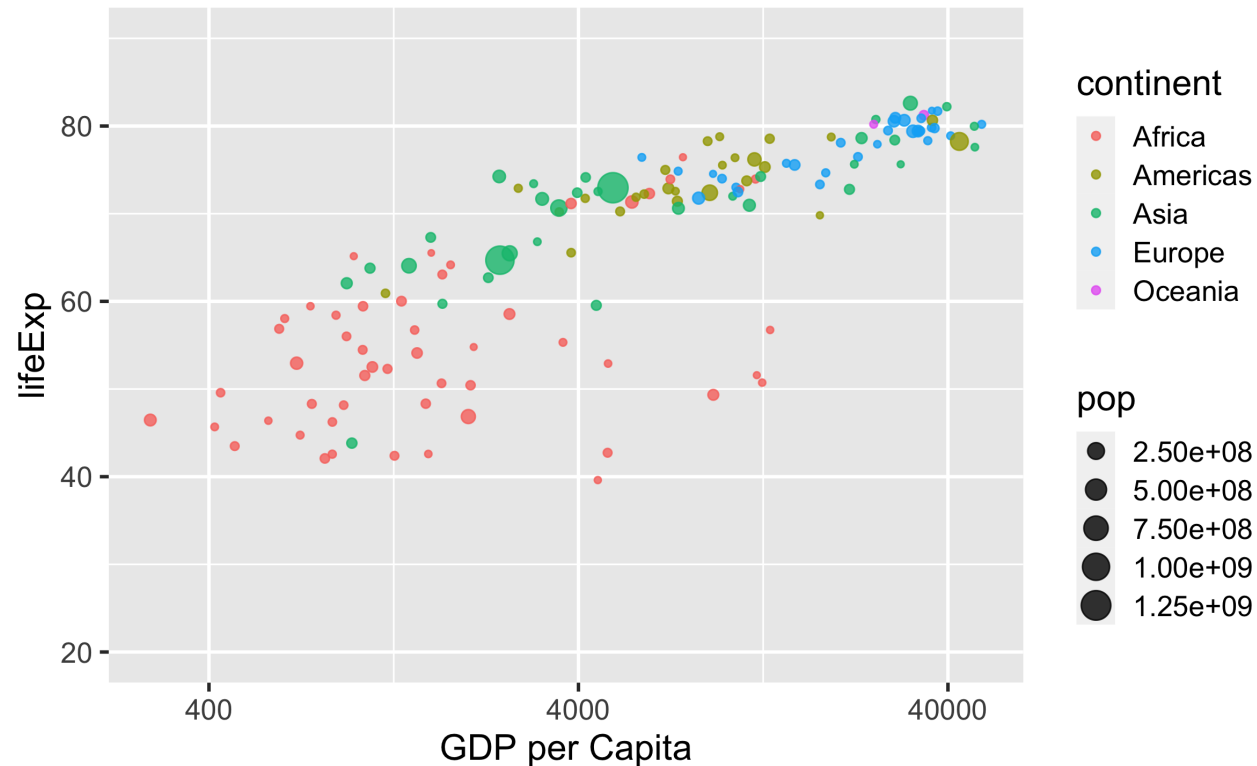
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPercap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop)
```



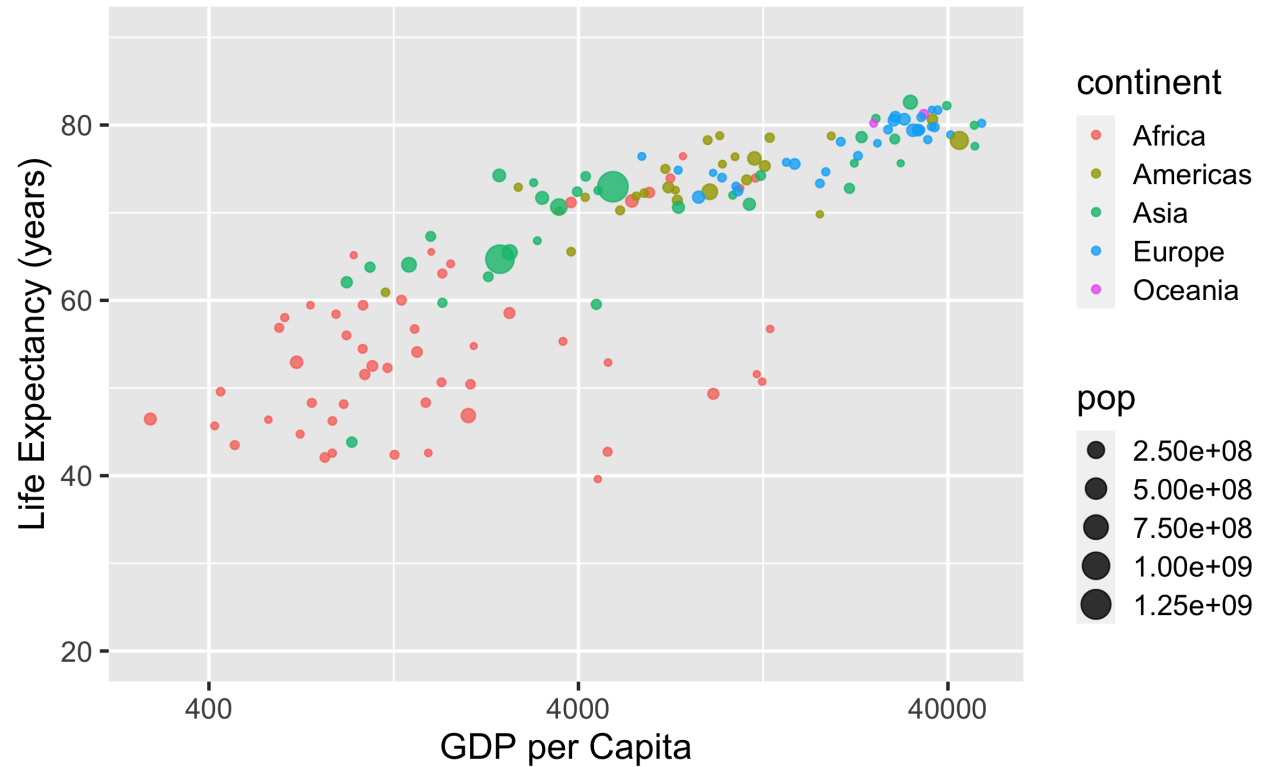
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPercap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10")
```



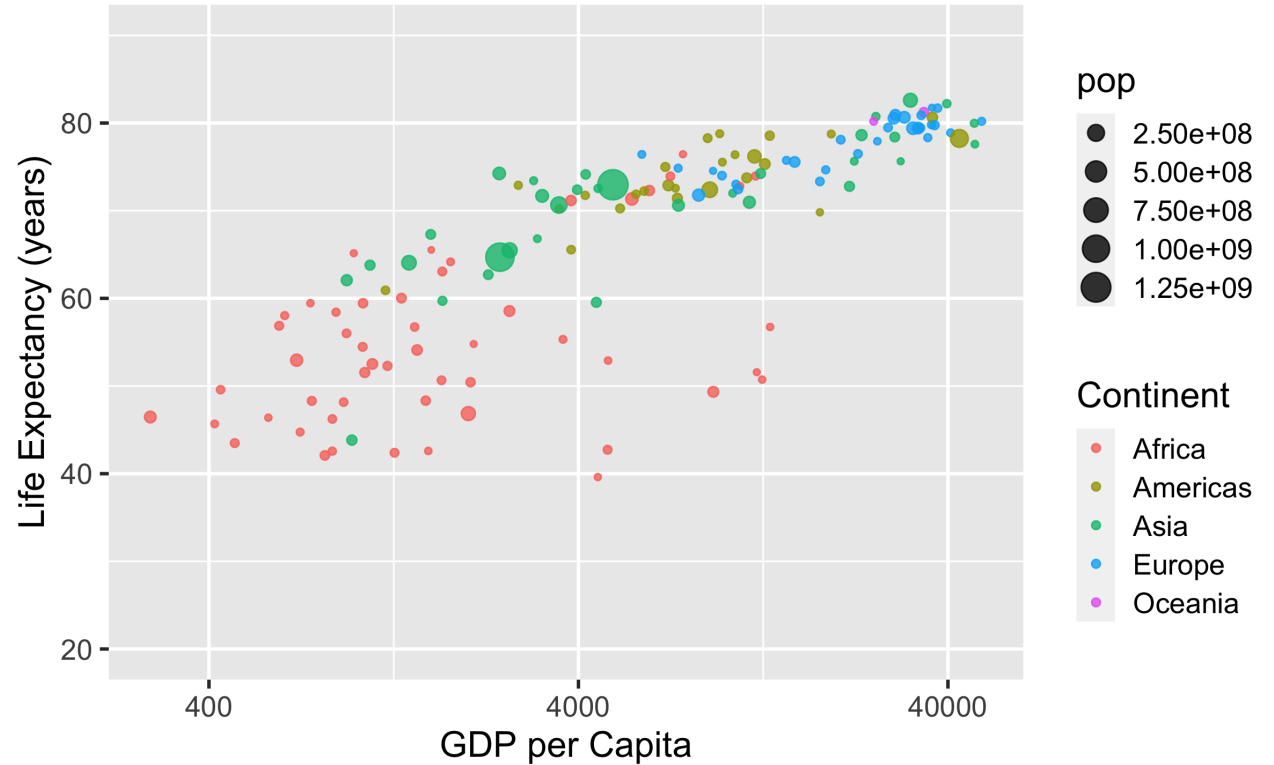
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPercap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita")
```



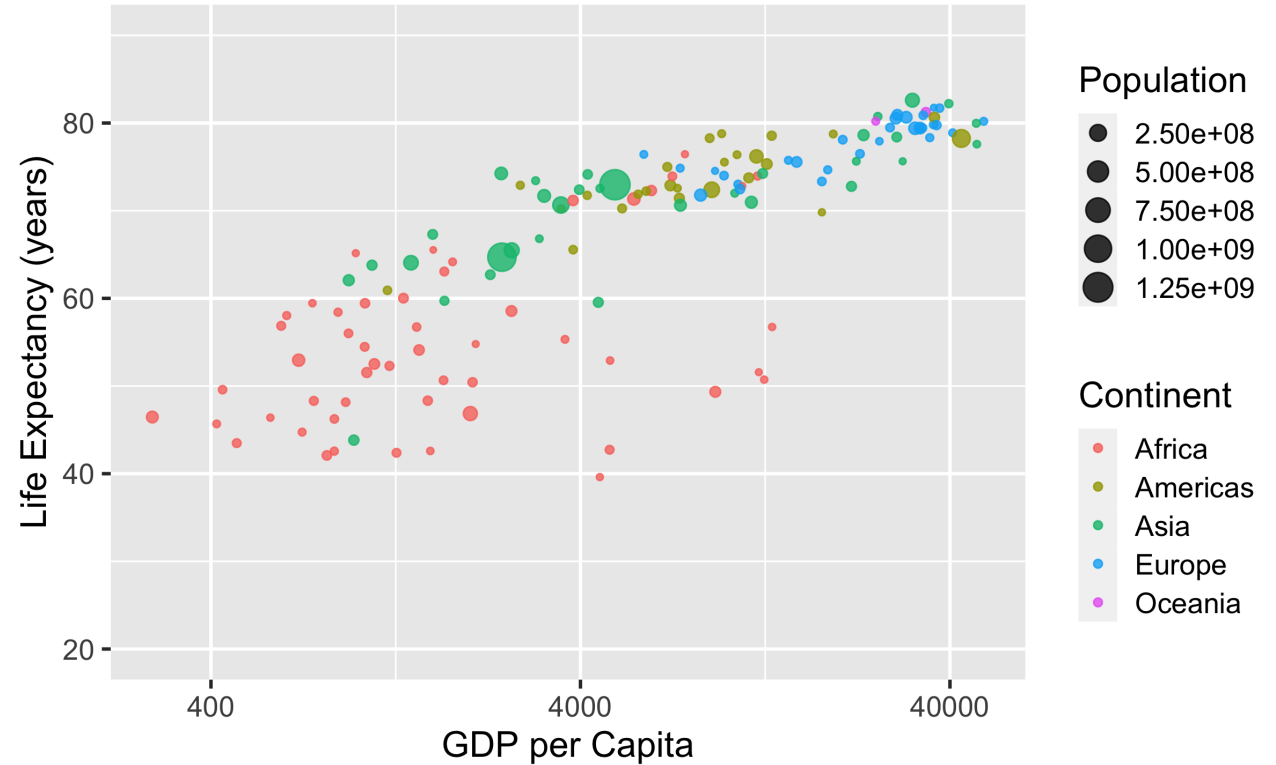
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPercap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)")
```



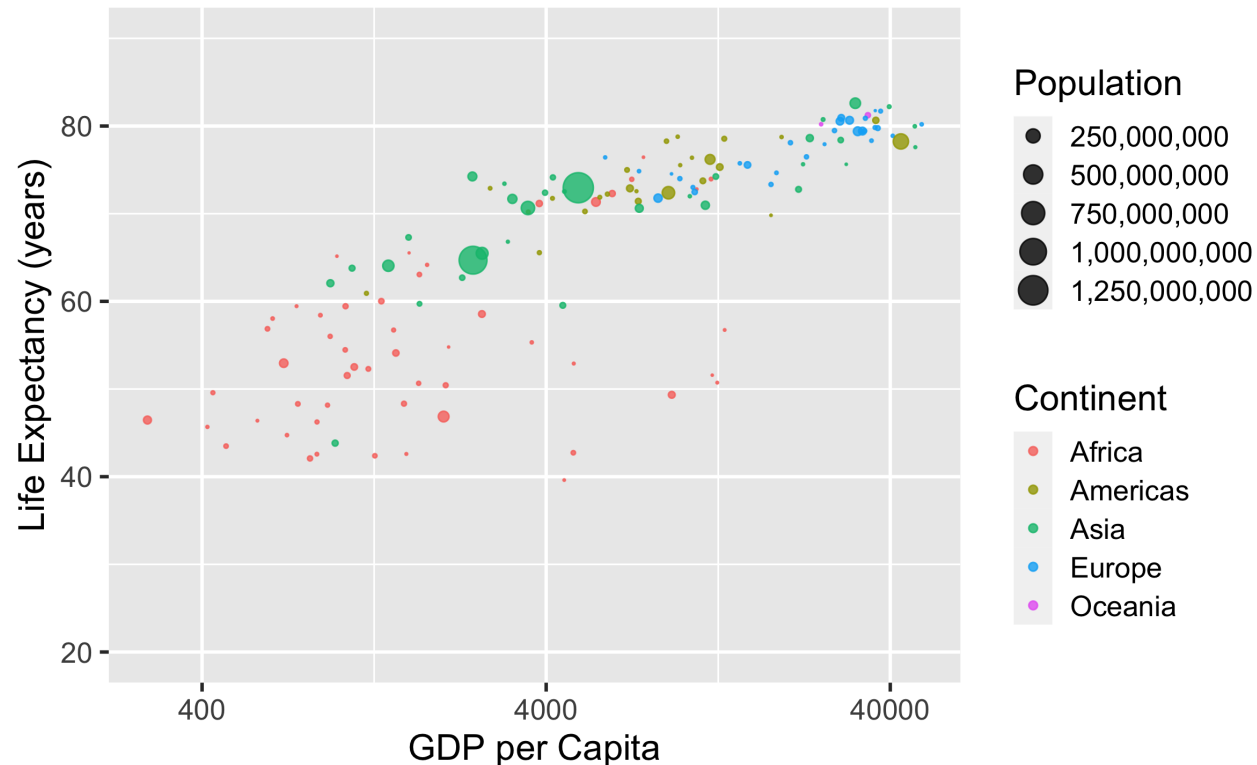
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPercap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)")
    labs(color = "Continent")
```



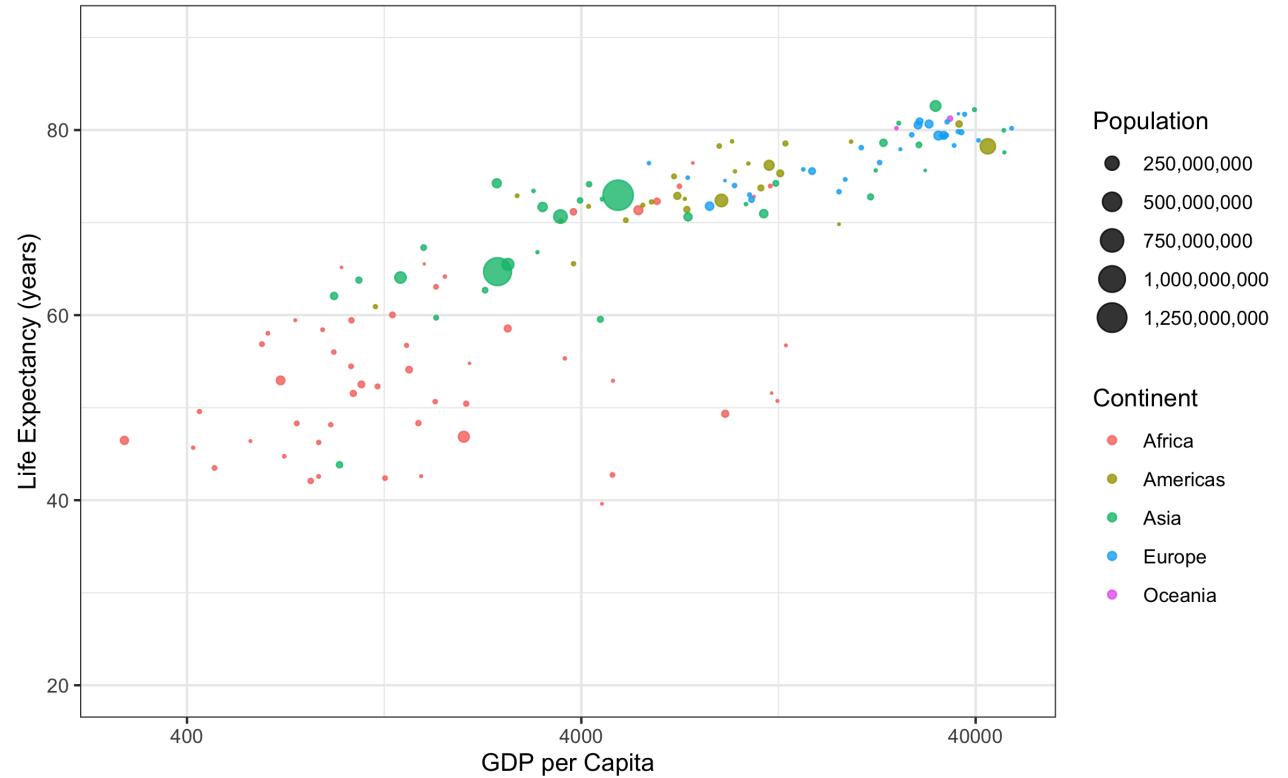
```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPerCap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)")
    labs(color = "Continent") +
    labs(size = "Population")
```



```
gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPerCap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)") +
    labs(color = "Continent") +
    labs(size = "Population") +
    scale_size_area(
      labels = label_comma())
```



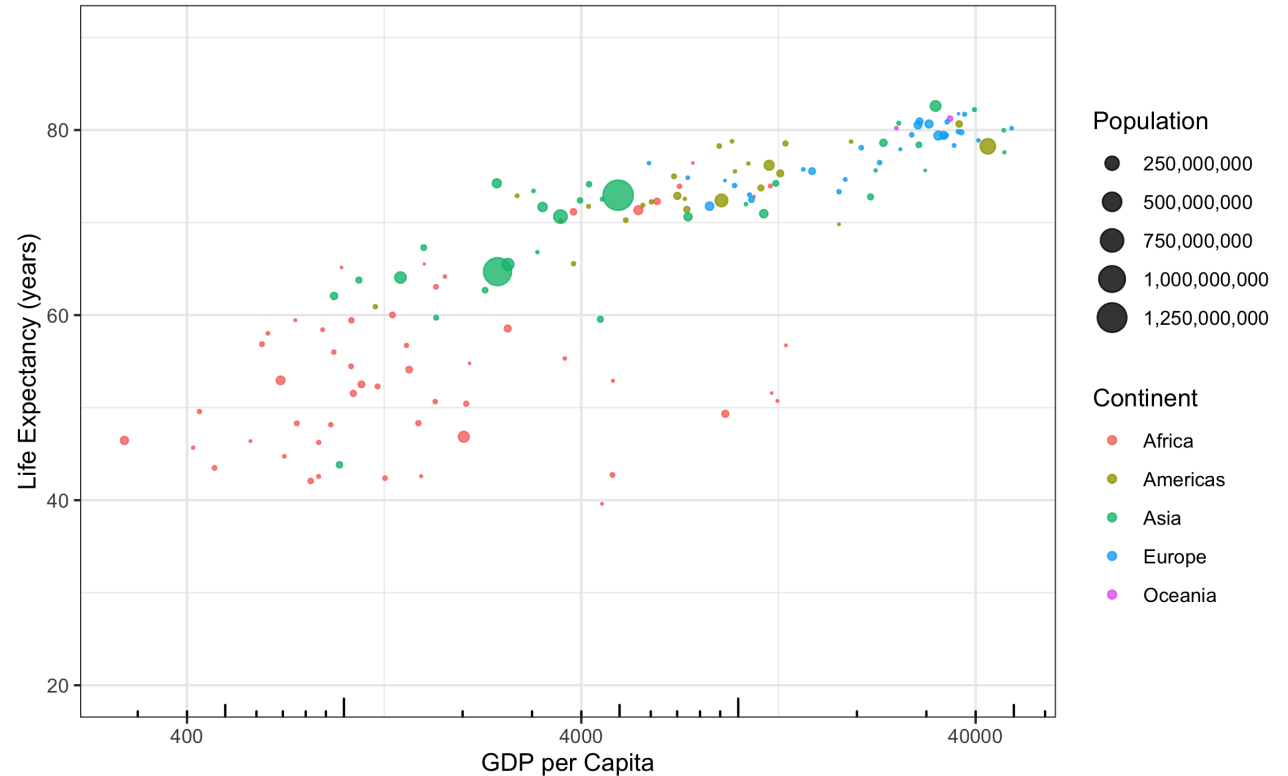
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gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
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    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)")
    labs(color = "Continent") +
    labs(size = "Population") +
    scale_size_area(
      labels = label_comma()) +
    theme_bw()
```



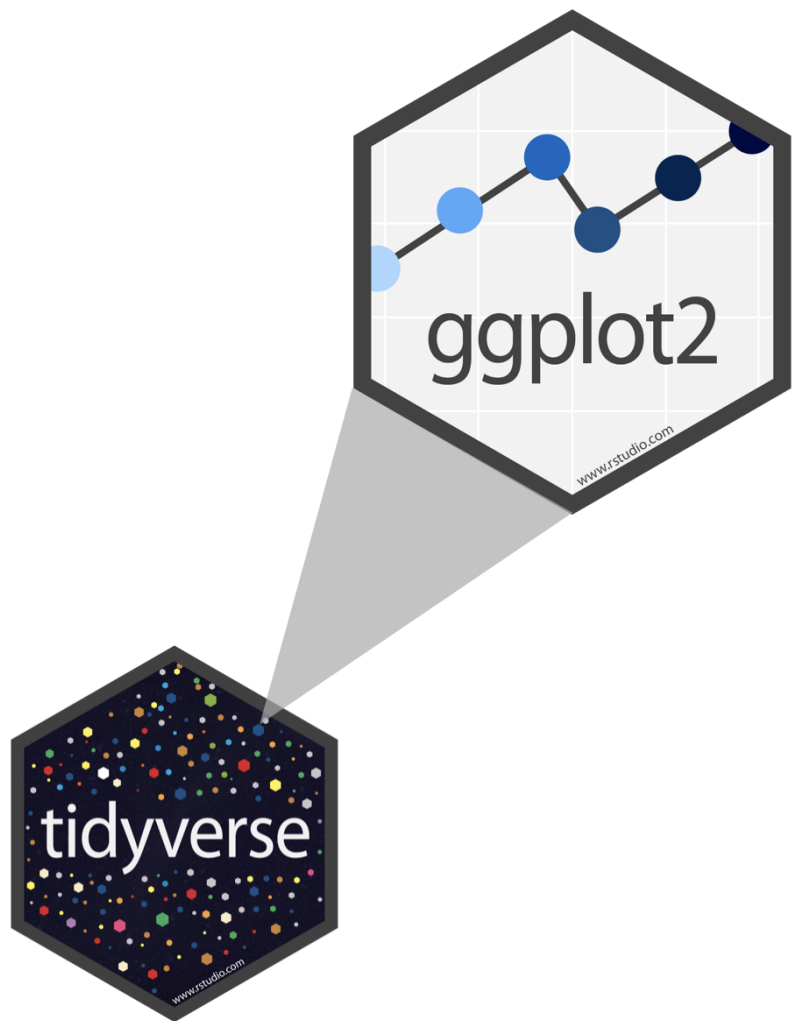
```

gapminder %>%
  filter(year == 2007) %>%
  ggplot() +
    aes(x = gdpPerCap, y = lifeExp) +
    geom_point(alpha = .8) +
    coord_cartesian(ylim = c(20, 90))
    aes(color = continent) +
    aes(size = pop) +
    scale_x_continuous(
      breaks = c(400, 4000, 40000),
      trans = "log10") +
    labs(x = "GDP per Capita") +
    labs(y = "Life Expectancy (years)")
    labs(color = "Continent") +
    labs(size = "Population") +
    scale_size_area(
      labels = label_comma()) +
    theme_bw() +
    annotation_logticks(sides = "b")

```



ggplot2 ∈ tidyverse

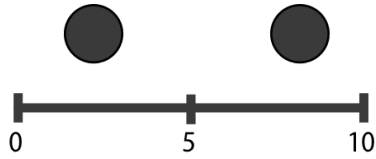


- **ggplot2** is tidyverse's data visualization package
- The **gg** in "ggplot2" stands for Grammar of Graphics
- It is inspired by the book **Grammar of Graphics** by Leland Wilkinson

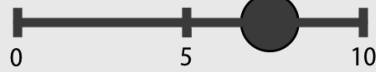
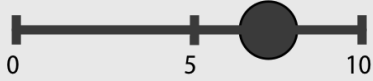
Grammar of Graphics

Concisely describe the components of a graphic





Position on
a common scale



Position on
unaligned
scales



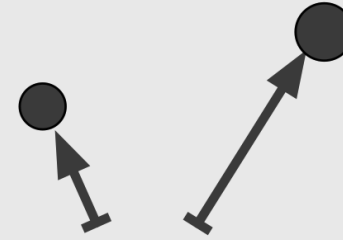
Length



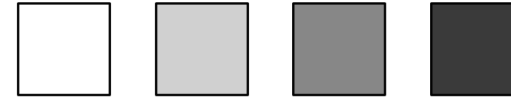
Tilt or Angle



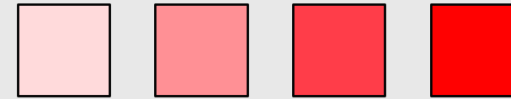
Area (2D as size)



Depth
(3D as position)



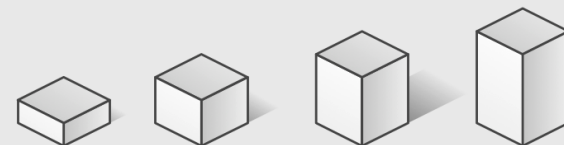
Color luminance
or brightness



Color saturation
or intensity



Curvature



Volume
(3D as size)

